



Chapter 6: Construction Process

6.1 Site Surveying and Preparation

- It is the **inspection** of an area where work is proposed, to gather information for a design or an estimate to complete the initial tasks required for an outdoor activity.
- It can determine **a precise location, access, best orientation** for the site and the location of obstacles'.
- The aim of survey is take all information from existing / new site to provide better solution and helpful network.
- The basic tools used for site survey is camera, GPS, compass, safety tools (e. g. belt, helmet, shoes, etc.)
- Procedure: Pre-survey > Preparation > Survey activity > Survey analysis > Design and solution > Implementation

Site Surveying and Preparation

- A good **site survey** will help to determine the feasibility of the desired coverage.
- The **process** of performing a site survey includes the following steps:
 - Gather tools and configuration
 - Research and investigate industry specific concerns
 - Gather the recommended equipment list / site survey kit
 - Implement the site survey
 - Document the site survey
- Before survey, all the related personnel are called to have **survey preparation coordination meeting** having the following objectives
 - Arranging survey and coordination personnel
 - Preparing vehicle and equipment
 - Making survey plan and confirming survey route
 - Offering the primary schemes of transmission and power

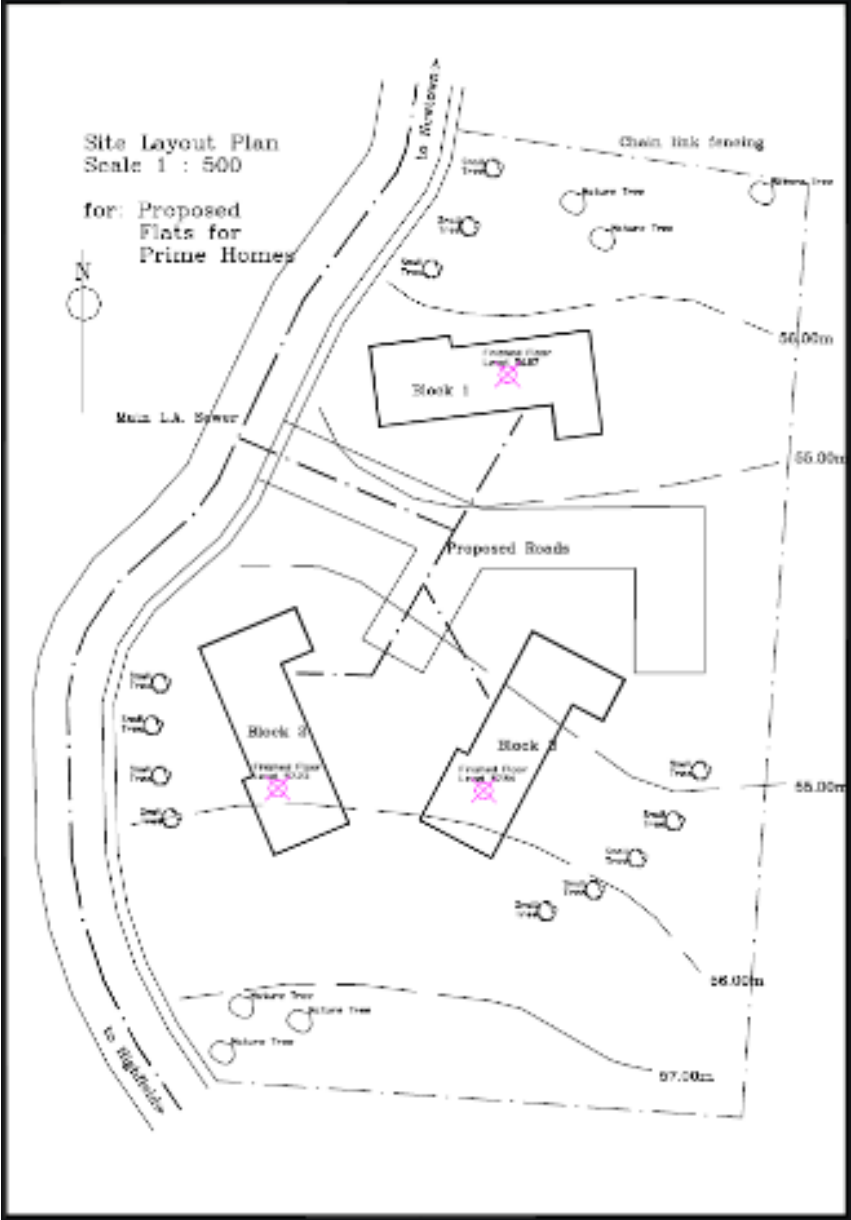


Site Surveying and Preparation

Setting Out

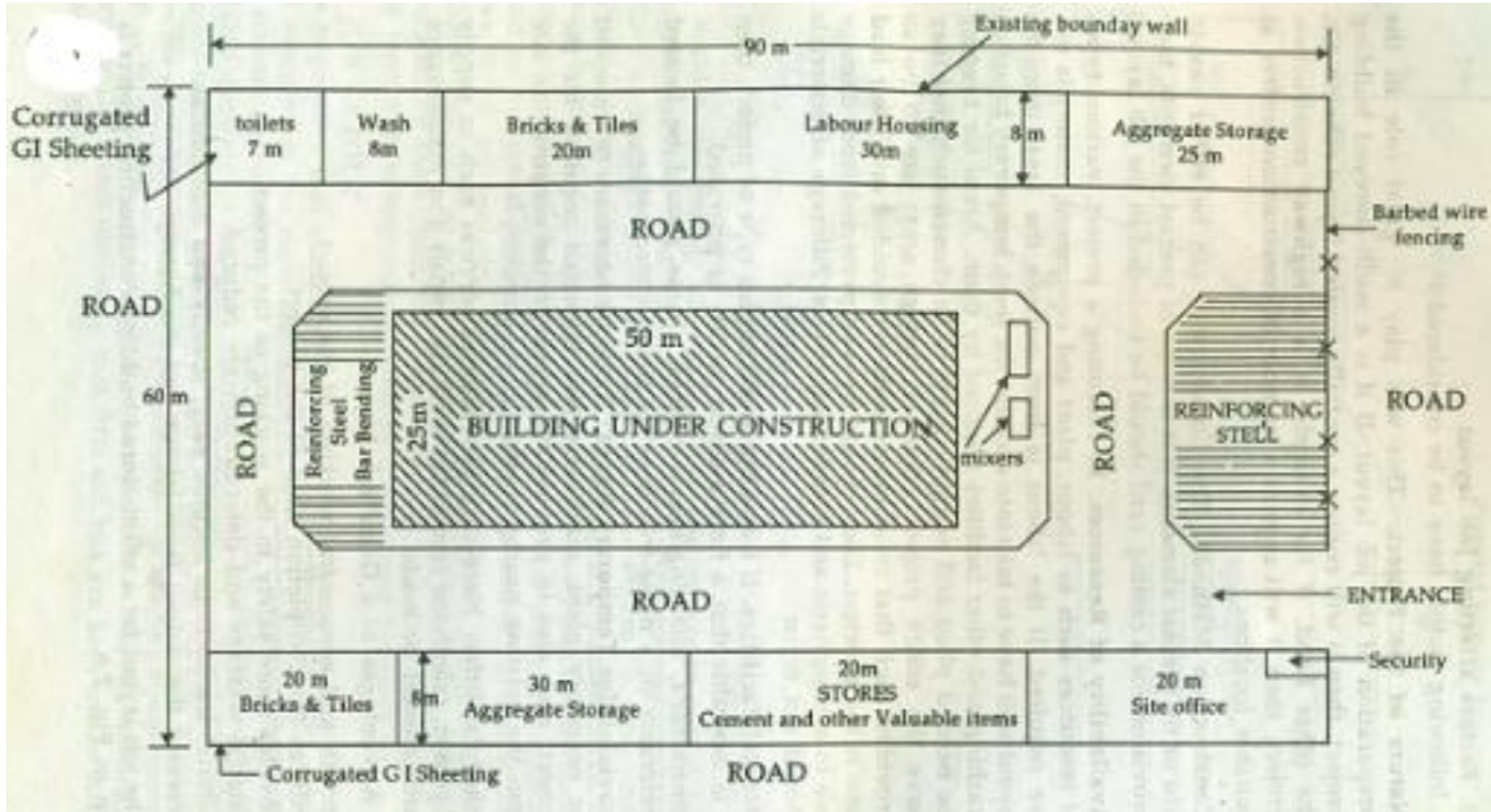
- To locate the point on the ground as per the drawing / location map with the help of reference point.
- We have to **assign the location** for major types of construction materials, equipment and facilities (e.g. cement, sand, gravel, brick, rebar, mixture, tower crane, other equipment, house for officer, house for labor, office house, guard house, toilet and gate etc.)
- Job/Site layout can be defined as site space allocation for material storage, working areas, units of accommodation, plant positions, general circulation areas and also access and egress for deliveries and emergency services.
- It is imperative that every construction project should be organized and executed in the most economical and safest manner.
- To aid on the smooth execution of the project, a job layout is prepared.
- This a scaled drawing of the proposed construction site showing all the relevant features such as entry and exit points to the site, storage areas for materials, contractors offices, area keeping for equipment such as mixer, bar bending area, labor housing, toilets etc.

Site Layout Plan



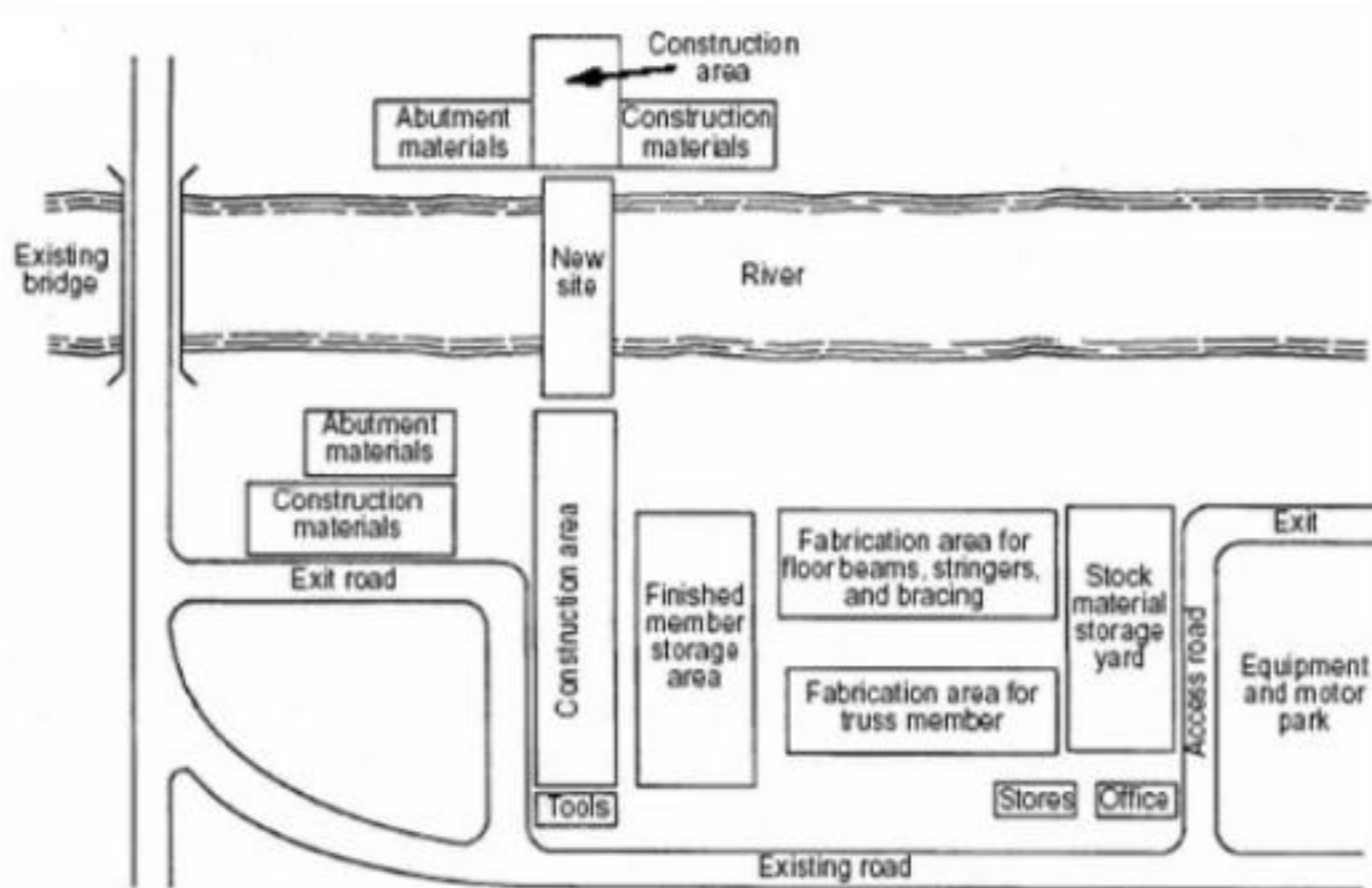
Site Layout Plan

Site Layout / Job Layout of a building project



Site Layout Plan

Construction Site Layout



6.2 Job Layout

- The larger and more complex the **project**, greater will be the need for planned **job layout** and detailing at **site**.
- Job/Site layout can be defined as site space allocation for material storage, working areas, units of accommodation, plant positions, general circulation areas and also access and egress for. **deliveries and emergency services**
- It is imperative that every **construction project** should be organized and executed in the **most economical and safest** manner.
- To aid on the **smooth execution** of the project, a job layout is prepared.
- This a **scaled drawing** of the proposed construction site showing all the relevant features such as entry and exit points to the site, storage areas for materials, contractors offices, area keeping for equipment such as mixer, bar bending area, labor housing, toilets etc.
- It is also necessity that **first aid kits are kept** in the site engineer's office.

Job Layout

The following factors should be considered in preparing site layout:

- **Nature, Scope and Type of work:** - the nature of site layout depends majorly on the type of work to be carried out, e. g. the site layout for a dam would be much more complicated and extensive as compared to that of a residential building.
- **Topography of the site:** - the topography of the site i.e. location, size, nature of terrain etc. of the site also effect the site layout. e. g. the site layout in case of rocky mountains terrain would be vastly different from that in case of a plain terrain.
- **Methods adopted for execution of work:** - the method selected for the execution of work deeply affects the nature of site layout e.g. if the work is carried out mostly mechanically i.e. using more equipment and less labor, more emphasis has to be placed on providing facilities for the machinery than for the workers.

Job Layout

The following are the advantages of a good job layout: -

- Smooth and economical working of the project.
- Reduces the completion time of the project.
- Provides more safety in the working of the project.
- Material wastage and deterioration are reduced.
- Material transactions become easy, speedy and economical.
- Increases the output from labor and machinery.

6.3 Material Handling

- Material handling is the **movement, protection, storage and control** of materials and products throughout manufacturing, warehousing, distribution, consumption and disposal.
- It **establishes a plan** which includes basic requirements, desirable alternates and planning for contingency.
- It **integrates handling and storage** activities, which is cost effective into integrated system design.

Material Handling

Advantages of Effective Material Handling

- Improve customer service
- Reduce inventory excesses
- Shorten delivery times
- Lower overall handling costs in manufacturing, distribution and transportation
- Enhance handling through streamlining and better timing
- Facilitate product control through better monitoring and management
- Improve inventory management and the accuracy of information through real-time monitoring
- **Least handling is the best handling**
- Research has shown that best handling of construction materials can **reduce cost by 4 to 5 % of total cost.**

6.4 Financial Management & Cash Flow Management

Financial Management

- A **construction financial** manager's background often indicates the areas in which the manager will concentrate.
- For example, a **construction financial** manager whose background is in **construction** operations (estimating and **project management**) initially will concentrate on the proper recording of job costs.
- Up to submission of tender; contractors have to spend in **buying tender documents** (Rs.1000 to 10,0000), **field visit, preparing bid /tender** (approximately 1% of bid amount) and bid security (2.5 % of bid amount)
- After selection; certain amount is required for signing contract ad performance security deposit.
- Contractor may get certain amount (about 20% of contract amount) as mobilization fund.

Financial Management/ Cash Flow

Financial Issues:

- Lack of financial facilities restrict growth of construction industry
- Lengthy and cumbersome disbursement and payment procedures lead to project delay.

Management Deficiencies:

- Some contractors and consultants lack professional management.
- Inadequate research and development.
- Lack of capacity in Project Managers.
- A cash flow forecast (also called a 'cash flow budget' or 'cash flow projection') helps identify whether a firm needs to borrow, how much, when, and how it will repay the loan.
- Building a cash flow forecast allows an evaluation of cash resources that are required and when they are required.

Project Cash Flow:

- **Project cash flow** is the net **cash flow** associated with the **project** for that year.
- **Project Cash Flow** = Sources of **Cash** – Uses of **Cash**.
- **Project** Financial Planning provides a way to indicate the **cash flow** incidence for the account or **project**.

Project Cash Flow Diagram

- Project cash flow diagrams are used as a tool to visualize the costs of an investment.
- In cash flow diagrams, x-axis is time and y-axis is the money that is made or lost during a time period.
- Typically money that is being invested or paid out are drawn pointing down from the time axis and money that is being returned is drawn pointing up.

Tips For Managing Cash Flows On Construction Projects

- Know your client
- Start with a realistic and profitable estimate
- Perform a cash flow forecast or projection
- Negotiate favorable contract terms
- Be aggressive in collecting payments
- Stay on top of change orders
- Close out the project

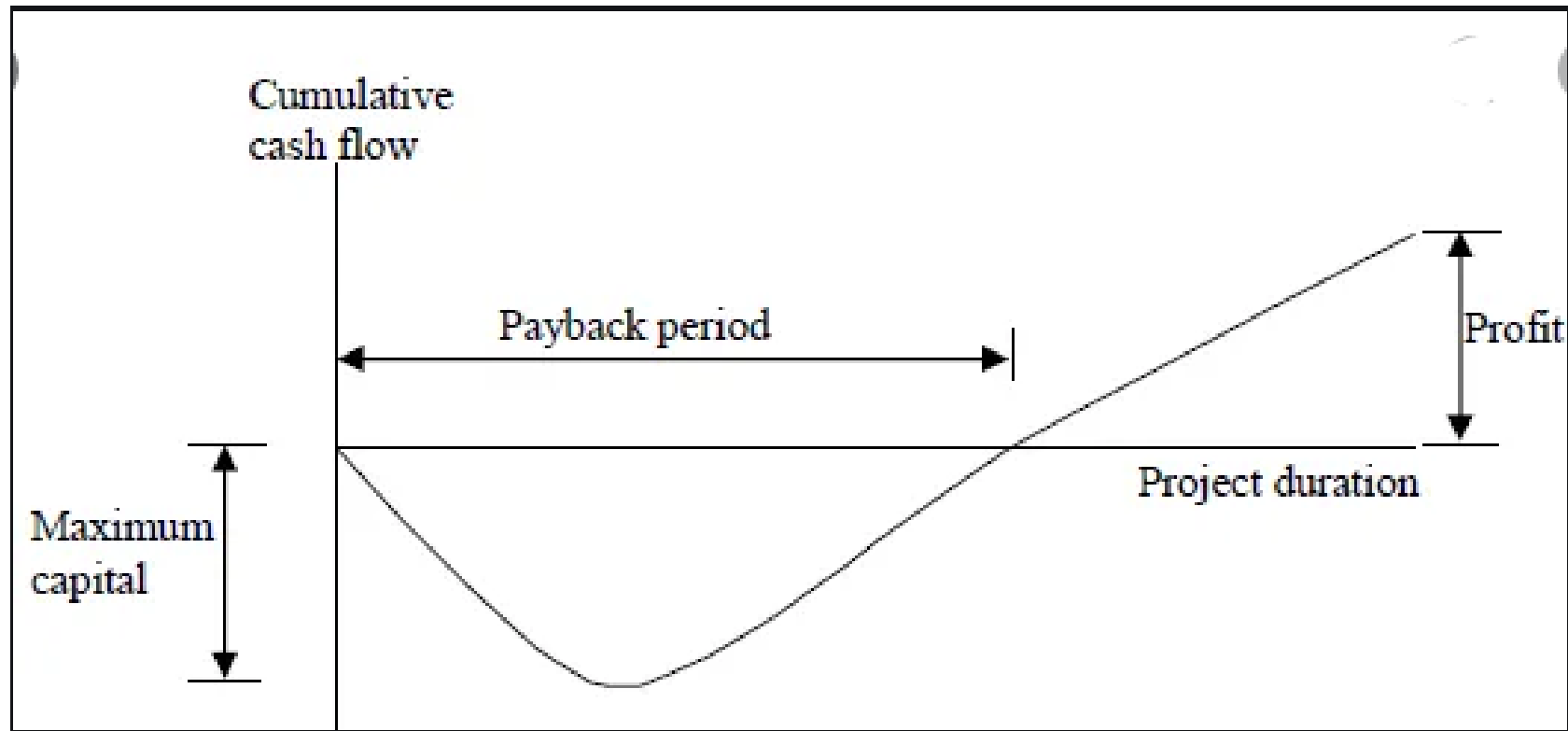
Project Cash Flow

Best practices for managing a **healthy cash flow** are as below:

- 1) Identify business risks & prepare in advance
 - 2) Create a separate bank account for your business
 - 3) Monitor your inventory efficiently
 - 4) Always keep buffer money
 - 5) Implement better system to manage cash flow
 - 6) Control cash outflows
 - 7) Keep your cash growing
 - 8) Do not focus on profit, focus on cash flow. If your cash flow is in order, your profit will be in order.
- Without cash, profits are meaningless.
 - Many a profitable business on paper has ended up in **bankruptcy** because the amount of cash coming in doesn't compare with the amount of cash going out.
 - Firms that don't exercise good cash management may not be able to make the investments needed to compete, or they may have to pay more to borrow money to function.

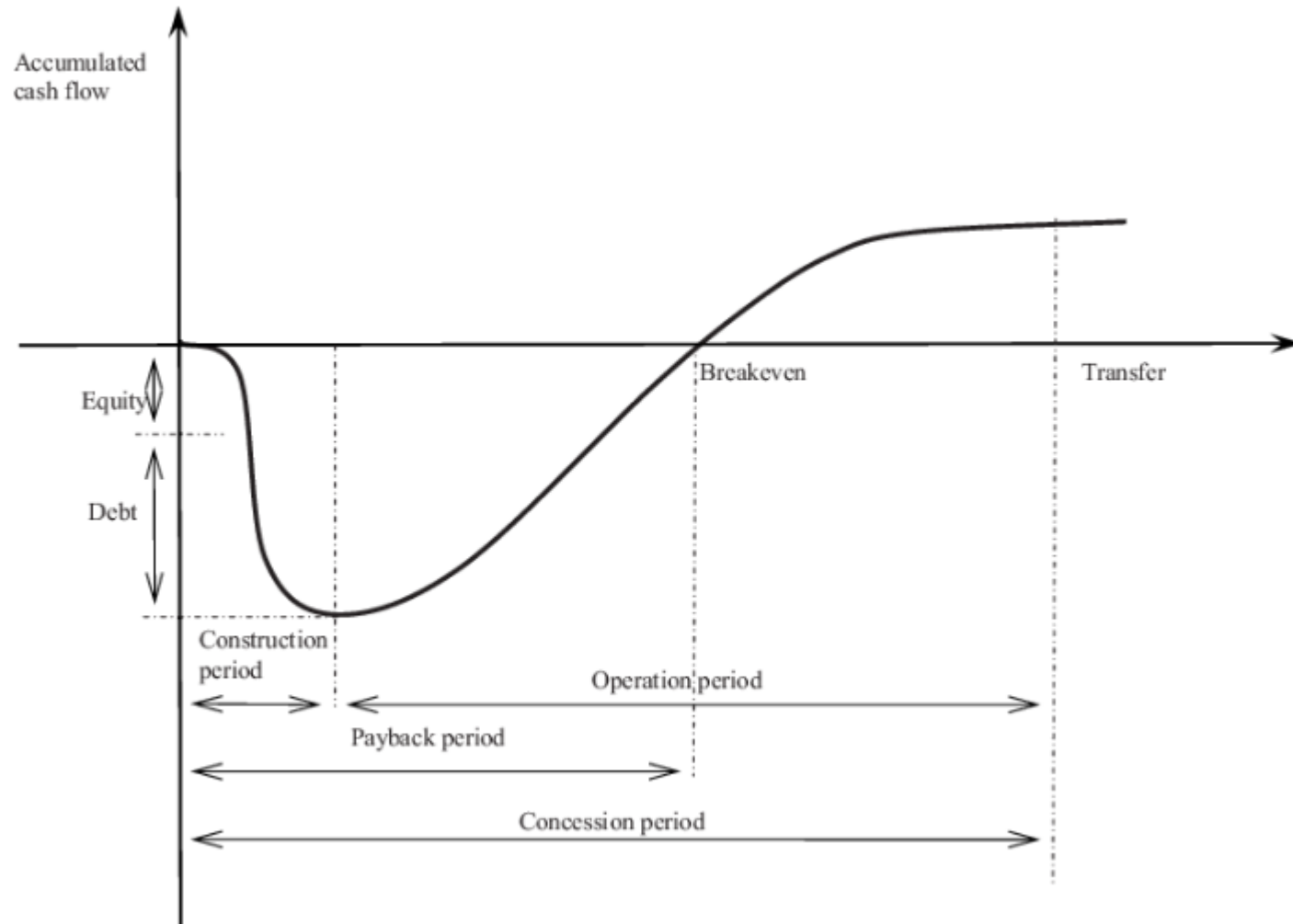
Project Cash Flow Diagram

Project Cash Flow



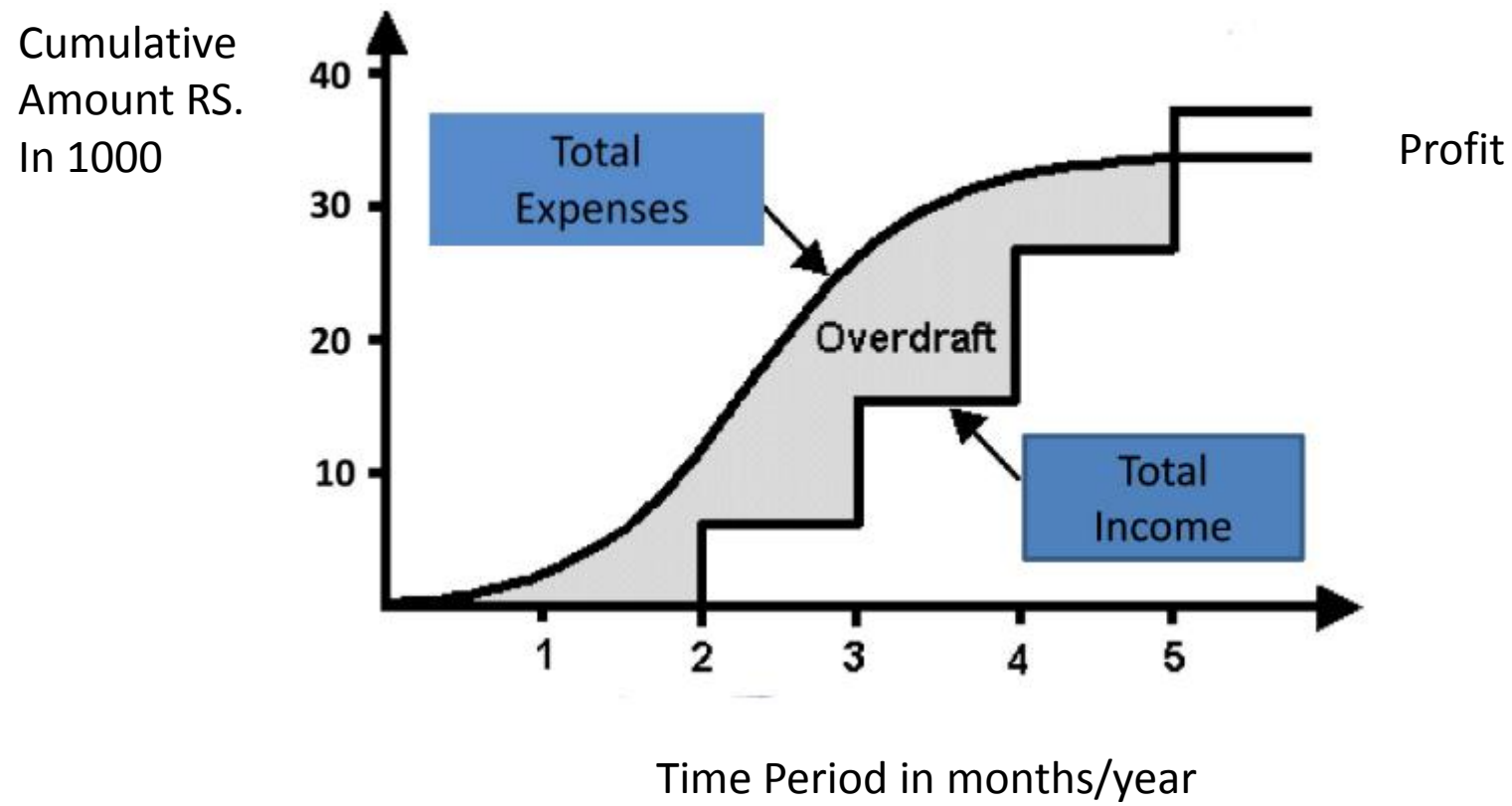
Project Cash Flow Diagram

Typical Cash Flow Profile of BOT Project



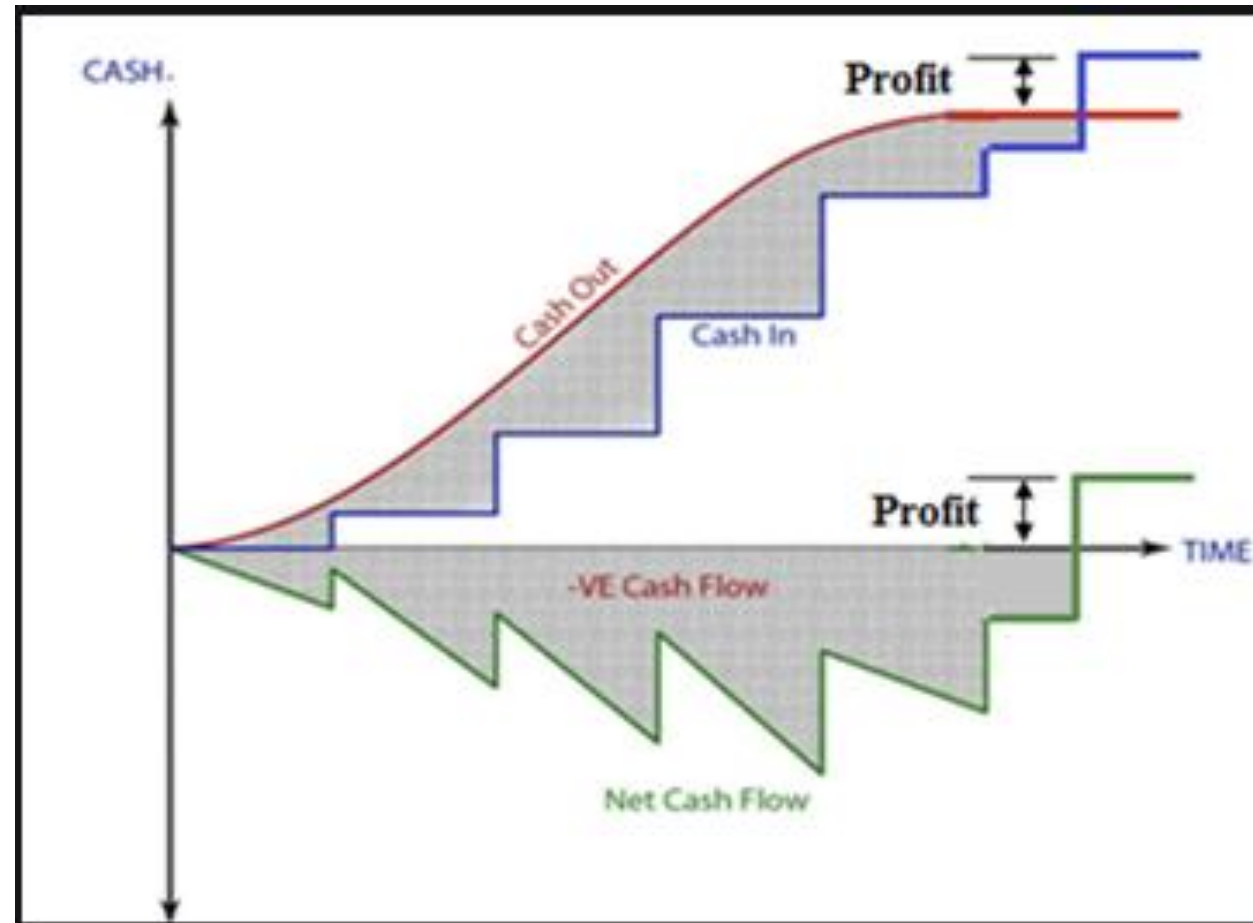
Project Cash Flow Diagram

Real Construction Cash Flow (if no mobilization cost)



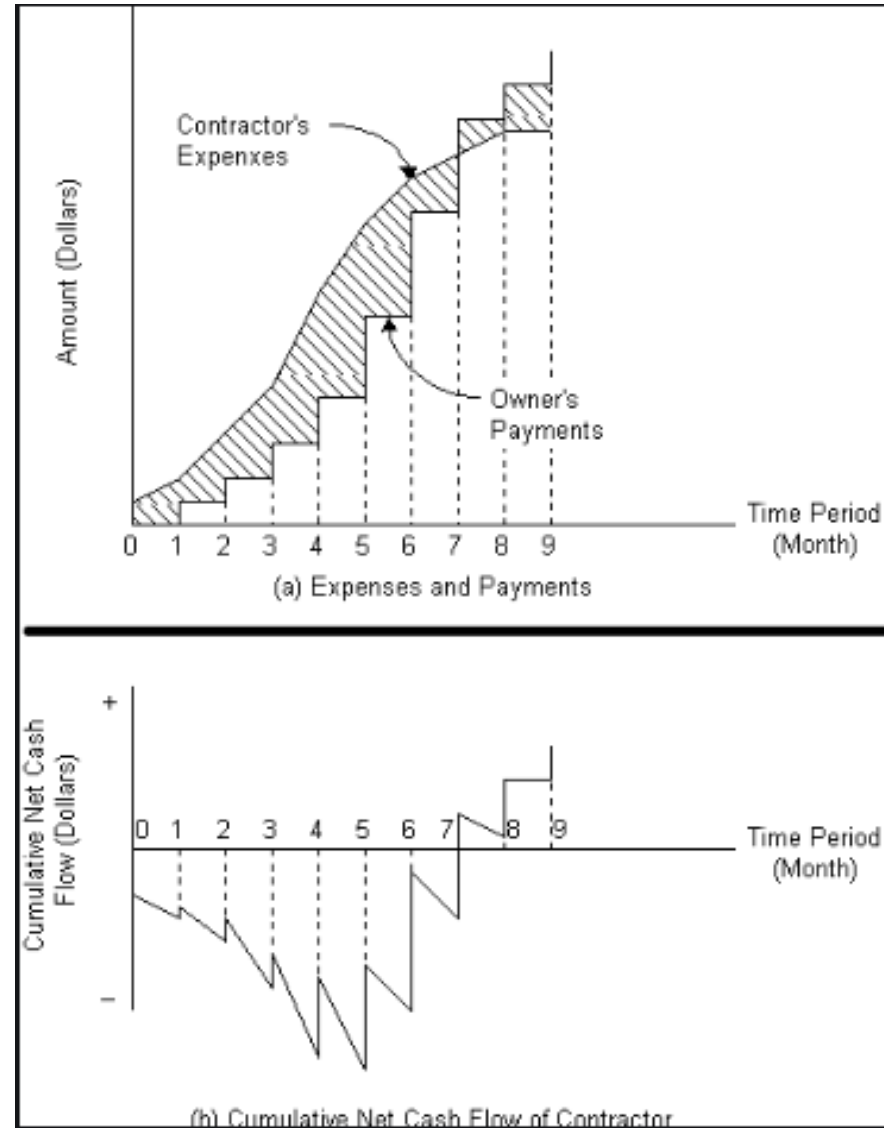
Project Cash Flow Diagram

Net Cash Flow (if no mobilization cost)



Project Cash Flow Diagram

Cash Flow



Project Cash Flow Diagram

Project Cash Flow Diagram (no mobilization fund / mobilization fund available)

